Solution Exam2 Practice

1. $\frac{e-1}{2}\ln 2$
2. $-\frac{1}{\pi}\sin(\pi^2)+\pi$
3. $1/2(e-1)$
4. $\sin(2/3)$
5. $\frac{1}{4}(e^{16}-e)$
6. $\pi/2$
7. $\pi/3$
8. $\frac{\pi}{2}(1-\cos(16))$
9. 0
10. $(-1/2)(\cos(1)-\cos(0))$
11. $1/4\sin(1)(e-1)$
12. $(2/9)(e-2)$
13. $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{36-x^2-y^2}} 1\, dzdydx$
14. $\int_0^{2\pi}\int_0^1\int_r^{\sqrt{36-r^2}} r\, dzdrd\theta$
15. $\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^1(\rho\cos(\phi))^2\rho^2\sin(\phi)d\rho d\phi d\theta$
16. $\int_0^{1/\sqrt{2}}\int_0^{2\pi}\int_z^{\sqrt{1-z^2}} z^2 r dr d\theta dz$ or $\int_0^{2\pi}\int_0^{1/\sqrt{2}}\int_0^r z^2 r dz dr d\theta+\int_0^{2\pi}\int_{1/\sqrt{2}}^1\int_0^{\sqrt{1-r^2}} z^2 r dz dr d\theta$
17. (a) $\int_0^{2\pi}\int_0^{\pi/4}\int_0^{3\sec\phi}\rho^2\sin(\phi)d\rho d\phi d\theta$

    (b) $\int_0^{2\pi}\int_0^3\int_r^3 r dz dr d\theta$

    (c) $\int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}\int_{\sqrt{x^2+y^2}}^{3} dzdydx$

18. (a) $\int_0^{2\pi}\int_0^{\pi/3}\int_{\sec\phi}^{2}\rho^2\sin(\phi)d\rho d\phi d\theta$
    (b) $\int_0^{2\pi}\int_0^{\sqrt{3}}\int_1^{\sqrt{4-r^2}} rdzdrd\theta$
    (c) $\int_{-\sqrt{3}}^{\sqrt{3}}\int_{-\sqrt{3-x^2}}^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} dzdydx$

19. (a) $\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^{\csc\phi}\rho^2\sin(\phi)d\rho d\phi s\theta$
    (b) $\int_0^{2\pi}\int_0^{1}\int_0^{r} rdzdrd\theta$
    (c) $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_0^{\sqrt{x^2+y^2}} dzdydx$

20. $\int_0^{2\pi}\int_0^{1/\sqrt{2}}\int_r^{\sqrt{1-r^2}} rdzdrd\theta$

21. $\int_0^{2\pi}\int_0^{\pi/2}\int_0^{1}\rho^2\sin\phi d\rho d\phi d\theta$

22. $\int_1^5\int_0^{2\pi}\int_0^{z} rdrd\theta dz$

23. $\int_{-1}^{1}\int_0^{2\pi}\int_0^{\sqrt{9-z^2}} rdrd\theta dz$

24. $\int_{-1}^{0}\int_0^{2\pi}\int_0^{\sqrt{100-z^2}} r^3drd\theta dz$

25. $\int_0^{2\pi}\int_0^{2}\int_{-r}^{r} r^3\sin^2(\theta)\cos^2(\theta)dzdrd\theta$

26. $\int_0^{2\pi}\int_0^{2}\int_{r^2-6}^{8-2r^2} rdzdrd\theta$

27. $\int_0^{2\pi}\int_1^{2}\int_0^{r^2+1} rdzdrd\theta$

28. 0

29. -2/3

30. $\frac{\sqrt{2}}{3}\sin^3(\pi/2)$

31. $\pi/2+(1/3)(\pi/2)^3$

32. (a) $\int_0^1\sqrt{4t^2+25+e^{2t}}dt$; (b) $\int_0^1 t^4\sqrt{4t^2+25+e^{2t}}dt$

33. (a) $\int_0^{2\pi}\sqrt{4\cos^2(t)+4\sin^2(t)+1}dt = \int_0^{2\pi}\sqrt{4+1}dt = \sqrt{5}\int_0^{2\pi}dt$; (b) $\sqrt{5}\int_0^{2\pi}(4-t)dt$

34. The parameterization is $x(t)=2\cos(t), y(t)=2\sin(t)$
    (a) $\int_0^{\pi}\sqrt{4\sin^2(t)+4\cos^2(t)}dt = 2\int_0^{\pi}dt = 2\pi$
    (b) $\int_0^{\pi}(2\cos(t))^2 2\sin(t)\sqrt{4\sin^2(t)+4\cos^2(t)}dt = 16\int_0^{\pi}\cos^2(t)\sin(t)dt$

35. $\int_0^5\langle t^2,2\rangle\cdot\langle 1,e^t\rangle dt = \int_0^5 t^2+2e^t dt = \frac{1}{3}t^3+2e^t|_0^5 = \frac{1}{3}(5)^3+2e^5-2$

36. $-2/3$
37. $1/3$
38. $1/5 + 1/2$
39. 0
40. 8π
41. 1
42. It is not conservative
43. $xy^2 + e^x + 1/2y^2$
44. It is not conservative
45. $ye^x + e^y$
46. $x^2e^{-y} + y^2$
47. $x\sin(y) + e^y$
48. $\frac{2}{3}$
49. 2
50. $\sin(1)$
51. Choices b, d, f, and g.